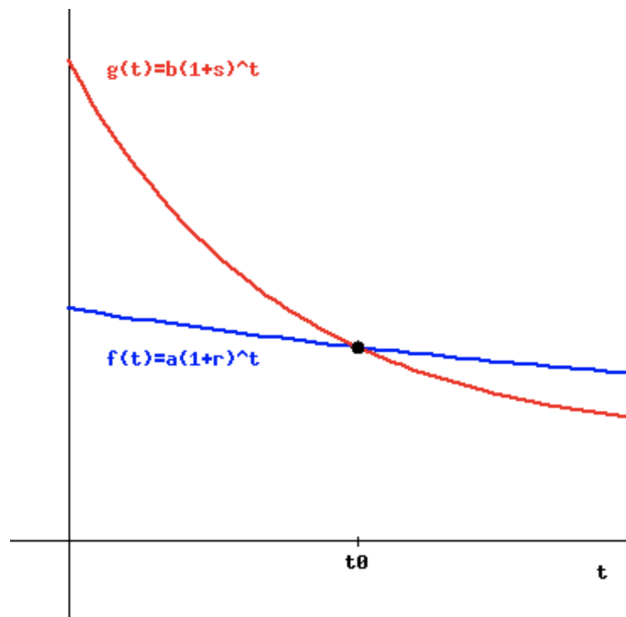


Chapter 3 Practice Assessment
Sections 3.1 - 3.4

1. Suppose $Q = 30.8(0.751)^t$. Give the starting value a , the growth factor b , and the growth rate r . In other words, determine a , b , and r such that $Q = ab^t = a(1+r)^t$.
2. (a) The annual inflation rate is 3.8% per year. If a movie ticket costs \$9.00 dollars today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.
(b) According to your formula, how much will a movie ticket cost in 30 years?
3. A population has size 8000 at time $t = 0$, with t in years.
(a) If the population decreases by 125 people per year, find a formula for $P(t)$, the population at time t .
(b) If the population decreases by 6% per year, find a formula for $P(t)$, the population at time t .
4. If $b > 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \rightarrow -\infty$? If there is no horizontal asymptote, say so.
5. Find the long term behavior of each of the following functions.
(a) As $x \rightarrow \infty$, $18(0.8)^x \rightarrow$
(b) As $t \rightarrow -\infty$, $9(2.2)^t \rightarrow$
(c) As $t \rightarrow \infty$, $0.6(2 + (0.1)^t) \rightarrow$
(d) As $t \rightarrow -\infty$, $11e^{0.14t} \rightarrow$
6. Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1+r)^t$ and $g(t) = b(1+s)^t$. If the value of r is increased and the values of a , b , and s remain the same, does the value of t_0 increase, decrease or stay the same?



7. Certain radioactive material decays in such a way that the mass remaining after t is given by the function

$$m(t) = 175e^{-0.015t}$$

where $m(t)$ is measured in grams.

- (a) Find the mass at time $t = 0$.
 - (b) How much of the mass remains after 15 years?
8. The graph of the function $f(x) = e^{-x} - 4$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations?
- (a) What are the transformation? Does the order in which you perform the transformations matter? If so, state the correct order.
 - (b) State the domain and range of $f(x)$.
9. Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.
- (a) $\ln 4 = x$
 - (b) $\ln x = 3$
10. Evaluate the following expressions. Your answers must be exact and in simplest form.
- (a) $\ln(e^9)$
 - (b) $e^{\ln 2}$
 - (c) $e^{\ln \sqrt{3}}$
 - (d) $\ln\left(\frac{1}{e^2}\right)$
11. Find the exact solution of the exponential equation $e^{2x+1} = 18$.
12. Find the exact solution of the logarithmic equation $6 - \ln(5 - x) = 0$.